Air University, Aerospace and Aviation Campus, Kamra
Department of Computer Science
BS. Cyber Security

DataBase Systems – Project Report
BSCYS – VI

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Executive Summary

This report presents the complete design and implementation of the Cyber Crime Reporting and Investigation Management System (CCRIMS), developed as a semester-end project for the Database Systems course. The system provides a centralized, database-driven platform that allows citizens to submit cyber crime complaints and enables investigation officers and forensic analysts to manage cases, track investigation progress, and maintain digital evidence.

The database is designed following BCNF normalization principles, resulting in 13 well-structured relational tables that eliminate data redundancy and maintain full referential integrity. Four user roles — Citizen, Officer, Analyst, and Admin — each have dedicated modules and access controls. The technology stack uses PHP 8.2, MySQL/MariaDB via XAMPP, and Tailwind CSS for the frontend.

DB Tables	User Roles	Crime Types	Normal Form
13 Tables	**4 Roles**	**8 Types**	**BCNF**

1. Introduction

1.1 Background

With the rapid expansion of internet usage across Pakistan and globally, cyber crimes have become one of the most prevalent and damaging forms of criminal activity. Offenses such as phishing, online fraud, identity theft, ransomware attacks, and cyber harassment are increasing in frequency and complexity, targeting both individual citizens and large organizations.

Traditional manual and paper-based systems for handling crime reports are wholly inadequate for managing cyber crime investigations. These systems suffer from data redundancy, poor coordination between departments, delayed information retrieval, and a lack of structured digital evidence management.

1.2 Problem Statement

The absence of a centralized information system makes it challenging for law enforcement to manage complaints, track investigations, and maintain digital evidence. Records are stored in separate files, leading to:

- Data redundancy and inconsistency across departments
- Delays in retrieving critical case information
- Difficulty coordinating between investigation officers and forensic analysts
- No structured audit trail for digital evidence and chain of custody
- Inability to monitor case status in real time

1.3 Objectives
CCRIMS provides:

- A citizen-facing complaint portal for reporting cyber crimes online
- A case management system for investigation officers to track and update cases
- A digital evidence repository with SHA-256 hash-based integrity verification
- A forensic reporting module for structured submission of expert findings
- A legal warrant management system for search and arrest warrants
- Role-based access control ensuring data is accessible only to authorized personnel

2. System Overview

2.1 User Roles

Role	Badge	Responsibilities
Citizen	No badge	Register account, file complaints, track complaint and case status in real time.
Officer	OFF-xxx	Manage assigned cases, add investigation logs, manage suspects, issue warrants.
Forensic Expert	FOR-xxx	Collect digital evidence with hash verification, submit forensic reports.
Admin	ADM-xxx	Full system access: staff management, all cases, complaints and oversight.

2.2 Core Modules

2.2.1 Citizen Portal
Citizens can register accounts and file complaints by selecting a cyber crime type and providing a detailed incident description. The system assigns a unique complaint ID and allows status tracking through to case closure.

2.2.2 Investigation Management
Officers manage cases from complaint approval through closure. They can update case status and priority, add timestamped investigation log entries with action types such as IP Trace, Domain Analysis, or Malware Analysis, and link suspects with their involvement type.

2.2.3 Digital Evidence Management

Forensic experts register evidence items per case. Each entry includes evidence type, file name, SHA-256 hash for integrity verification, file size, collection date, and a chain of custody narrative ensuring tamper-evidence and legal admissibility.

2.2.4 Forensic Reporting

Forensic experts submit structured reports linked directly to case records. Reports include report type, detailed findings, tools used, methodology, and submission date. A one-to-one relationship with case records ensures each record has at most one forensic report.

2.2.5 Warrant Management

Officers issue and track legal warrants associated with active cases. Four warrant types are supported: Search Warrant, Arrest Warrant, Data Disclosure Order, and Account Freeze. Each warrant records the issuing court, issue date, expiry date, and current status.

3. Entity Relationship Diagram (ERD)

3.1 Entities and Attributes

Entity	Primary Key	Key Attributes
CITIZEN	CitizenID (UUID)	Fname, Lname, Email, NationalID, Phone, City, Password
STAFF	StaffID (UUID)	Fname, Lname, Role, BadgeNumber, Department, Email, Password
CYBER_CRIME_TYPE	CrimeTypeID (UUID)	TypeName, Description
COMPLAINT	ComplaintID (UUID)	Description, LodgeDate, Status, CitizenID FK, CrimeTypeID FK
CASE	CaseID (UUID)	CaseNumber, Status, Priority, OpenDate, Jurisdiction, ComplaintID FK
CASE_STAFF	(CaseID + StaffID)	AssignedDate, AssignmentRole
SUSPECT	SuspectID (UUID)	Fname, Lname, Alias, Phone, City, NationalID, Status
CASE_SUSPECT	(CaseID + SuspectID)	InvolvementType
DIGITAL_EVIDENCE	EvidenceID (UUID)	EvidenceType, FileName, FileHash UNIQUE, FileSize, CaseID FK, CollectedBy FK

CASE_RECORD	RecordID (UUID)	RecordDate, Summary, CaseID FK, StaffID FK
INVESTIGATION_LOG	LogID (UUID)	ActionType, ActionDetail, Timestamp, RecordID FK, StaffID FK
FORENSIC_REPORT	RecordID (PK=FK)	ReportType, Findings, ToolsUsed, Methodology, ExpertID FK
WARRANT	WarrantID (UUID)	WarrantType, IssuedDate, ExpiryDate, IssuedBy, Status, CaseID FK

3.2 Relationships

Relationship	Cardinality	Description
CITIZEN — COMPLAINT	1 : N	One citizen can file many complaints
CYBER_CRIME_TYPE — COMPLAINT	1 : N	One crime type categorizes many complaints
COMPLAINT — CASE	1 : 1	Each complaint results in exactly one case
CASE — CASE_STAFF	M : N	Many staff members can work on many cases
CASE — CASE_SUSPECT	M : N	Many suspects can be linked to many cases
CASE — DIGITAL_EVIDENCE	1 : N	One case can have many evidence items
CASE — CASE_RECORD	1 : N	One case can have many case records
CASE_RECORD — FORENSIC_REPORT	1 : 1	Each case record produces at most one forensic report
CASE_RECORD — INVESTIGATION_LOG	1 : N	One case record can have many log entries
CASE — WARRANT	1 : N	One case can have multiple warrants issued
STAFF — DIGITAL_EVIDENCE	1 : N	One staff member collects many evidence items
STAFF — CASE_RECORD	1 : N	One staff member creates many case records

STAFF — FORENSIC_REPORT	1 : N	One forensic expert submits many reports

4. Enhanced ER Diagram (EERD)

The EERD extends the basic ERD with specialization, weak entities, and aggregation concepts specific to the CCRIMS schema.

4.1 Specialization — STAFF Entity

The STAFF entity undergoes disjoint total specialization into four subclasses based on the Role attribute. Every staff member belongs to exactly one role (disjoint) and all staff must belong to one of the four subclasses (total participation).

Subclass	Badge Prefix	Specialized Responsibilities
Officer	OFF-xxx	Conducts investigations, manages cases, issues warrants.
Analyst	ANL-xxx	Performs cyber intelligence and data analysis.
Forensic Expert	FOR-xxx	Collects digital evidence and submits forensic reports.
Admin	ADM-xxx	Manages all system data, staff accounts and oversight.

4.2 Weak Entity — CASE_RECORD

CASE_RECORD is a weak entity — it cannot exist without a parent CASE. Its partial key is (RecordDate + StaffID), which combined with the owning CaseID forms a unique identifier. If a CASE is deleted, all associated CASE_RECORDs are automatically removed via ON DELETE CASCADE.

4.3 Aggregation — FORENSIC_REPORT

FORENSIC_REPORT uses aggregation by treating the CASE_RECORD relationship as a higher-level abstraction. Rather than linking directly to CASE, the FORENSIC_REPORT is built on top of CASE_RECORD. This is reflected in the schema where RecordID serves as both the primary key and foreign key in FORENSIC_REPORT, enforcing the 1:1 relationship.

4.4 Multivalued Attributes — Resolved

In the original unnormalized design, CITIZEN and SUSPECT had multivalued Phone attributes (Phone1, Phone2, Phone3). These were resolved by merging a single Phone column directly into both tables, eliminating unnecessary junction tables since each entity requires at most one primary phone number.

4.5 Composite Key Entities

Two entities use composite primary keys, representing many-to-many junction relationships:

- CASE_STAFF — PK: (CaseID, StaffID) — links cases to staff members with assignment details
- CASE_SUSPECT — PK: (CaseID, SuspectID) — links cases to suspects with involvement type

5. Database Normalization

The CCRIMS database was designed through a systematic normalization process, progressing from an unnormalized raw structure through 1NF, 2NF, 3NF, and finally BCNF. Each stage eliminates a specific type of data anomaly to ensure a clean, efficient schema.

5.1 Unnormalized Form (0NF)

In the original manual system all data was stored in flat files. A single raw record contained:

```
CCRIMS_RAW(
  CitizenID, Fname, Lname, Email, Phone1, Phone2,
  ComplaintID, Description, LodgeDate, Status, CrimeTypeName,
  CaseNumber, CaseStatus, Priority, Jurisdiction,
  StaffID, StaffName, Role, BadgeNo,
  Evidence='log.txt, disk.img', SuspectPhone1, SuspectPhone2
)
```

5.2 First Normal Form (1NF)

Rule

Every column must be atomic. No repeating groups or arrays. Every table must have a primary key.

Violations Found and Fixes Applied

Violation	Before 1NF	After 1NF Fix
Repeating phone numbers (CITIZEN)	CITIZEN(..., Phone1, Phone2, Phone3)	CITIZEN(..., Phone) — single atomic column
Repeating phone numbers (SUSPECT)	SUSPECT(..., Phone1, Phone2)	SUSPECT(..., Phone) — single atomic column
Non-atomic address in SUSPECT	SUSPECT(..., '123 Main St, Karachi, Sindh')	Separate columns: Street, City each atomic

Multi-valued evidence list in CASE	CASE(..., Evidence='log.txt, disk.img')	DIGITAL_EVIDENCE table — one row per item
Bundled staff assignments	CASE(..., Officers='Ahmed, Sara, Bilal')	CASE_STAFF junction table with individual rows
Multiple actions in one record	CASE_RECORD(..., Actions='traced IP, seized device')	INVESTIGATION_LOG — one row per action

5.3 Second Normal Form (2NF)

Rule

Must be in 1NF. No partial dependencies — every non-key attribute must depend on the entire composite primary key.

Table Analyzed	2NF Analysis and Outcome
CASE_STAFF (CaseID, StaffID)	Role renamed to AssignmentRole to confirm it means the role played on this specific case. It depends on the full (CaseID, StaffID) composite key. No violation.
CASE_SUSPECT (CaseID, SuspectID)	InvolvementType describes how this suspect is involved in this case — depends on both keys together, not either one alone. No violation.
COMPLAINT	CrimeTypeName would partially depend on CrimeTypeID, not ComplaintID. Fixed by keeping CYBER_CRIME_TYPE as a separate table referenced via FK.
Phone junction tables (removed)	CITIZEN_PHONE and SUSPECT_PHONE had composite PKs. Since only one phone is needed per entity, these were merged into parent tables, eliminating unnecessary junction tables.

5.4 Third Normal Form (3NF)

Rule

Must be in 2NF. No transitive dependencies — non-key attributes must not depend on other non-key attributes.

Table / Scenario	3NF Analysis
CITIZEN / SUSPECT — Address	City stored as a direct atomic column. No ZipCode column was added, thus avoiding the ZipCode -> City -> State transitive dependency chain entirely.

STAFF — Candidate Keys	StaffID, BadgeNumber, and Email are all candidate keys (UNIQUE). Since all three are superkeys, no transitive dependency exists between them.
CASE — 1:1 with COMPLAINT	ComplaintID declared UNIQUE in CASE, making it a candidate key. No transitive dependency through the FK.
FORENSIC_REPORT	RecordID is both PK and FK (1:1 with CASE_RECORD). All attributes depend directly on RecordID. No non-key attribute determines another.
DIGITAL_EVIDENCE — FileHash	FileHash is UNIQUE (candidate key / superkey). FileHash -> FileName is valid since FileHash is a superkey. No 3NF violation.

5.5 Boyce-Codd Normal Form (BCNF)

Rule

For every non-trivial functional dependency X -> Y, X must be a superkey. Stricter than 3NF — catches any non-superkey determinant.

Table	BCNF Status	Assessment
CITIZEN	**PASS**	Email and NationalID declared UNIQUE (candidate keys). All FDs have superkeys as determinants.
STAFF	**PASS**	StaffID, BadgeNumber, Email all superkeys. Every FD has a superkey determinant. No violation.
CYBER_CRIME_TYPE	**PASS**	TypeName declared UNIQUE. Both CrimeTypeID and TypeName are candidate keys.
COMPLAINT	**PASS**	ComplaintID is the sole PK. No non-superkey determinants exist among non-key attributes.
CASE	**PASS**	ComplaintID UNIQUE (2nd candidate key). CaseNumber UNIQUE. All determinants are superkeys.
CASE_STAFF	**PASS**	Composite PK (CaseID, StaffID) is sole candidate key. All attributes depend on full key.
CASE_SUSPECT	**PASS**	Composite PK (CaseID, SuspectID). InvolvementType depends on full key. No violation.
SUSPECT	**PASS**	NationalID declared UNIQUE making it a candidate key alongside SuspectID.

DIGITAL_EVIDENCE	PASS	FileHash UNIQUE. FileHash -> FileName is valid since FileHash is a superkey.
CASE_RECORD	PASS	RecordID sole PK. Optional UNIQUE on (CaseID, StaffID) enforces one record per pair.
INVESTIGATION_LOG	PASS	LogID sole PK. All attributes depend on LogID only. No non-superkey determinant.
FORENSIC_REPORT	PASS	RecordID as PK. 1:1 with CASE_RECORD. All FDs have superkey determinants.
WARRANT	PASS	WarrantID sole PK. Status ENUM constrained. No BCNF violation.
MALWARE (removed)	N/A	Hash -> MalwareName was a potential BCNF issue. Resolved by removing table and merging into DIGITAL_EVIDENCE.EvidenceType.

5.6 Normalization Summary

Table	1NF Fix	2NF Fix	3NF Fix	BCNF Fix
CITIZEN	Phone atomic column	—	—	UNIQUE on Email, NationalID
STAFF	—	—	—	UNIQUE on BadgeNumber, Email
COMPLAINT	PKs added, atomic fields	CrimeType in separate table	—	—
CASE	—	—	ComplaintID UNIQUE (1:1)	CaseNumber UNIQUE declared
CASE_STAFF	Separated from CASE	Role -> AssignmentRole	—	—
SUSPECT	Phone atomic, PKs added	—	City stored directly	UNIQUE on NationalID
DIGITAL_EVIDENCE	Separated from CASE	—	FileHash UNIQUE	—

CASE_RECORD	Atomized action list	—	—	—
FORENSIC_REPORT	Separated from CASE_RECORD	—	RecordID as PK (1:1)	—
INVESTIGATION_LOG	Atomized log entries	—	—	—
WARRANT	—	—	—	Status ENUM constrained
MALWARE (removed)	Was sub-table	Hash candidate key	Hash->Name transitive	Removed, merged into EvidenceType

6. Database Schema

The complete SQL schema for all 13 BCNF-normalized tables. UUIDs (CHAR(36)) are used as primary keys. All foreign keys include ON DELETE CASCADE where appropriate.

6.1 Schema Overview

Table	PK Type	Foreign Keys	Purpose
CITIZEN	UUID	—	Stores citizen/victim accounts
STAFF	UUID	—	Stores all staff accounts by role
CYBER_CRIME_TYPE	UUID	—	Lookup table for crime categories
COMPLAINT	UUID	CitizenID, CrimeTypeID	Records submitted crime complaints
CASE	UUID	ComplaintID (UNIQUE)	Active investigation cases
CASE_STAFF	Composite	CaseID, StaffID	M:N assignment of staff to cases
SUSPECT	UUID	—	Records of identified suspects
CASE_SUSPECT	Composite	CaseID, SuspectID	M:N linking suspects to cases

DIGITAL_EVIDENCE	UUID	CaseID, CollectedBy	Digital evidence items per case
CASE_RECORD	UUID	CaseID, StaffID	Official investigation activity log
INVESTIGATION_LOG	UUID	RecordID, StaffID	Detailed action log per record
FORENSIC_REPORT	RecordID (=FK)	RecordID, ExpertID	Expert forensic analysis report
WARRANT	UUID	CaseID	Legal warrants linked to cases

6.2 Key Table Definitions

CITIZEN

```
CREATE TABLE CITIZEN (
  CitizenID  CHAR(36)     NOT NULL DEFAULT (UUID()) PRIMARY KEY,
  Fname      VARCHAR(50)  NOT NULL,
  Lname      VARCHAR(50)  NOT NULL,
  Email      VARCHAR(150) NOT NULL UNIQUE,
  NationalID VARCHAR(50)  NOT NULL UNIQUE,
  Phone      VARCHAR(20),
  City       VARCHAR(80),
  Password   VARCHAR(255) NOT NULL
);
```

COMPLAINT

```
CREATE TABLE COMPLAINT (
  ComplaintID CHAR(36) NOT NULL DEFAULT (UUID()) PRIMARY KEY,
  Description TEXT     NOT NULL,
  LodgeDate   DATE     NOT NULL DEFAULT (CURRENT_DATE),
  Status      ENUM('Pending','Under Review','Resolved','Rejected') DEFAULT 'Pending',
  CitizenID   CHAR(36) NOT NULL,
  CrimeTypeID CHAR(36) NOT NULL,
  FOREIGN KEY (CitizenID)   REFERENCES CITIZEN(CitizenID),
  FOREIGN KEY (CrimeTypeID) REFERENCES CYBER_CRIME_TYPE(CrimeTypeID)
);
```

CASE

```
CREATE TABLE `CASE` (
  CaseID      CHAR(36)    NOT NULL DEFAULT (UUID()) PRIMARY KEY,
  CaseNumber  VARCHAR(50) NOT NULL UNIQUE,
  Status      ENUM('Open','Active','Closed','Suspended') DEFAULT 'Open',
  Priority    ENUM('Low','Medium','High','Critical')     DEFAULT 'Medium',
  OpenDate    DATE        NOT NULL DEFAULT (CURRENT_DATE),
  CloseDate   DATE,
```

```
  Jurisdiction VARCHAR(150),
  ComplaintID  CHAR(36)    NOT NULL UNIQUE,
  FOREIGN KEY (ComplaintID) REFERENCES COMPLAINT(ComplaintID)
);
```

DIGITAL_EVIDENCE

```
CREATE TABLE DIGITAL_EVIDENCE (
  EvidenceID     CHAR(36)     NOT NULL DEFAULT (UUID()) PRIMARY KEY,
  EvidenceType   VARCHAR(100) NOT NULL,
  FileName       VARCHAR(255) NOT NULL,
  FileHash       VARCHAR(255) NOT NULL UNIQUE,
  FileSize       VARCHAR(50),
  CollectedDate  DATE         NOT NULL DEFAULT (CURRENT_DATE),
  ChainOfCustody TEXT,
  CaseID         CHAR(36)     NOT NULL,
  CollectedBy    CHAR(36)     NOT NULL,
  FOREIGN KEY (CaseID)      REFERENCES `CASE`(CaseID) ON DELETE CASCADE,
  FOREIGN KEY (CollectedBy) REFERENCES STAFF(StaffID)
);
```

FORENSIC_REPORT

```
CREATE TABLE FORENSIC_REPORT (
  RecordID      CHAR(36)     NOT NULL PRIMARY KEY,
  ReportType    VARCHAR(100) NOT NULL,
  Findings      TEXT,
  ToolsUsed     TEXT,
  Methodology   TEXT,
  SubmittedDate DATE         NOT NULL DEFAULT (CURRENT_DATE),
  ExpertID      CHAR(36)     NOT NULL,
  FOREIGN KEY (RecordID) REFERENCES CASE_RECORD(RecordID) ON DELETE CASCADE,
  FOREIGN KEY (ExpertID) REFERENCES STAFF(StaffID)
);
```

WARRANT

```
CREATE TABLE WARRANT (
  WarrantID   CHAR(36)     NOT NULL DEFAULT (UUID()) PRIMARY KEY,
  WarrantType VARCHAR(100) NOT NULL,
  IssuedDate  DATE         NOT NULL,
  ExpiryDate  DATE,
  IssuedBy    VARCHAR(150) NOT NULL,
  Status      ENUM('Active','Expired','Executed','Revoked') DEFAULT 'Active',
  CaseID      CHAR(36)     NOT NULL,
  FOREIGN KEY (CaseID) REFERENCES `CASE`(CaseID) ON DELETE CASCADE
);
```

7. Technology Stack

CCRIMS runs on a LAMP-style stack via XAMPP on Windows. All components are free and open-source.

Layer	Technology	Role in CCRIMS
Frontend	HTML5 + CSS3 + Tailwind CSS + JavaScript	Responsive dashboards for all four roles. Session-driven navigation and role-aware access control.
Backend	PHP 8.2 (Procedural)	Server-side form processing, session management, authentication, and database queries via PDO.
Database	MySQL / MariaDB 10.4	Stores all 13 normalized tables. Enforces FK constraints, UUID keys, ENUM types, and UNIQUE indexes.
Server	Apache 2.4 via XAMPP	Hosts the PHP application locally on Windows. Project folder is in htdocs, accessed via localhost.
DB Admin	phpMyAdmin	Database administration, SQL import, table browsing and query execution via browser interface.

8. Conclusion

CCRIMS successfully centralizes all cyber crime data into a single relational database with 13 BCNF-normalized tables. The system eliminates data redundancy, ensures referential integrity, and provides a structured platform for the full lifecycle of a cyber crime investigation — from initial complaint through investigation, evidence collection, forensic reporting, and legal warrant management.

The normalization process progressing through 1NF, 2NF, 3NF, and BCNF ensures every table is free from partial dependencies, transitive dependencies, and non-superkey determinants. Key design decisions such as UUID primary keys, UNIQUE constraints on candidate keys, ON DELETE CASCADE on weak entity relationships, and the 1:1 RecordID primary-foreign key in FORENSIC_REPORT all reflect a carefully considered database design.

8.1 Key Achievements

- 13 BCNF-normalized tables with full referential integrity and zero data redundancy
- Complete citizen-to-case pipeline: complaint filing through investigation to case closure
- Digital evidence management with SHA-256 hash and chain of custody tracking
- Structured forensic reporting with tools, methodology, and expert identification
- Legal warrant tracking with type, authority, issue date, expiry, and status

- Role-based access control for four distinct user types
- 20 sample rows per table providing realistic Pakistani cyber crime test data

8.2 Future Enhancements

- Real-time case status notifications via email or SMS to citizens
- Cross-case malware fingerprint matching using the FileHash field
- Statistical dashboard with crime type trends and case resolution rates
- Integration with NADRA database for real-time CNIC verification
- Mobile application for citizen complaint filing

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